Valerio Tarditi

BIOMEDICAL ENGINEERING

CONTACTS

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24 July 2000

Driver license B

PROFILE

Biomedical Engineering graduate specialized in Biomedical Instrumentation, with a keen interest in biomechanics and medical engineering.
Experienced in musculoskeletal simulation and wearable assistive device development.
Eager to contribute to innovation in medical devices and rehabilitation technologies.

LANGUAGES

Italian: Native
English: B2 (Cambridge First Certificate)

SOFT SKILLS

- Teamwork
- Scientific writing
- Problem solving
- Critical thinking
- Continuous learning

EDUCATION

Master's Degree in Biomedical Engineering

Politecnico di Torino, Turin — 2023 - 2026

- Biomedical Instrumentation specialization.
- Thesis: "Simulation based design and musculoskeletal modeling of FeatherExo for hip-extension assistance during gait".
- Final grade: 110/110 *cum laude*

Bachelor's Degree in Biomedical Engineering

Politecnico di Torino, Turin — 2019 - 2023

- Final Project: "Application of denoising algorithms on fluorescence images for GANs".
- Final grade: 105/110

High School Diploma

Liceo Scientifico "Maria Curie", Pinerolo (TO) — 2014 - 2019

- Final grade: 100/100 *cum laude*

EXPERIENCES

Guest Student

Istituto Italiano di Tecnologia (IIT)
Rehab Technologies, Genoa — Nov 2025 - July 2026

Contributed to the simulation-based design and validation of a lower-limb exoskeleton for Parkinson's disease patients, working at the intersection of mechanical engineering, biomechanics, and computational modelling.

- Built a high-fidelity digital twin of the human-exoskeleton system (OpenSim), integrating hardware kinematics and compliant transmission mechanics;
- Formulated and solved optimal control problems (NLP) to simulate and predict human-machine interaction under real-use conditions;
- Quantified device performance across three biomechanical metrics: mechanical unloading, neuromotor adaptation, and metabolic relief, providing data-driven design feedback.

Curricular Intern

Politecnico di Torino
Turin — Sept 2022 - March 2023

Utilized a MATLAB GUI to apply BM3D and CLAHE denoising algorithms on fluorescence microscopy images (cell clones, cardiospheres), pre-processing datasets for GAN training

TECHNICAL SKILLS

- Programming: MATLAB, Python
- Software: Blender (basic level), Microsoft Office, Meshlab, Mokka, OpenSim, Solidworks, Unity 6 (basic level).